Changjiang Han

M.Sc. Student in Machine Learning, MBZUAI
Researcher, Cyber-Physical Intelligence Lab (CPIL)

Machine Learning Department, MBZUAI, Abu Dhabi, United Arab Emirates
Email Homepage GitHub LinkedIn Google Scholar



Research Interests

My research builds LLM agents that are capable, personalized, and learn reliably from feedback.

Agentic Reinforcement Learning: how agentic RL data scales across tasks, environments and trajectories.
Agent Memory: turning retrieved knowledge into parametric memory that needs no online retrieval.
LLM Personalization: the internal mechanisms that drive personalization in LLMs, and how to steer them.
Generative Recommendation: semantic-ID recommenders, from collaborative signals to deployment.
Multi-Agent Systems: what agents communicate, and how to audit and evaluate it.

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — Abu Dhabi, UAE
M.Sc., Machine Learning Department — Sept. 2025 – Present
GPA: 4.00/4.00. Supervisor: Prof. Xue (Steve) Liu.

Huazhong University of Science and Technology (HUST) — Wuhan, China
B.Eng. in Electronic and Information Engineering (Elite Class) — Sept. 2021 – June 2025
GPA: 3.88/4.00 (6/29).

Publications & Patents

* equal contribution.

First and Co-First Author

Changjiang Han, Bowei He, Ye Yuan, Weixu Zhang, Haolun Wu, Xue Liu, Jikun Kang. *From Lookups to Lessons: Turning Retrieved Knowledge into Parametric Memory*. (**ICLR 2027**, under review)

Changjiang Han, Qingyang Li, Yaqiang Zang, Jikun Kang, Pinghua Gong, Xue Liu, Bowei He. *Restoring Collaborative Signals in Semantic-ID Generative Recommendation via Personalized Natural Language*. (**WWW 2027**, under review)

Guo Tang*, Hanye Wu*, **Changjiang Han***, Qingyang Li, Ming Zhang, Xiangyu Qian, Yanchen Qiao, Huanjie Wang, Zhi Ma, Zhen Li, Yaqiang Zang, Pinghua Gong. *OxygenREC-v2: Internalizing Discrimination into Generative Recommendation*. (**KDD 2027**, under review)

Changjiang Han, et al. *An Automated Chemical Material Preparation Platform with Dual Collaborative Robotic Arms*. Chinese Invention Patent Application No. 202411034150.4, 2024.

Co-Authored

Bowei He, Yankai Chen, Xiaokun Zhang, **Changjiang Han**, Ye Yuan, Chong Li, Xue Liu. *How Data Scales in Agentic Reinforcement Learning: Laws and Synthesis Strategies*. In **NeurIPS 2026**.

Ye Yuan, Rui Song, Weien Li, Zeyu Li, Haochen Liu, Xiangyu Kong, **Changjiang Han**, Yonghan Yang, Zichen Zhao, Zixuan Dong, Fuyuan Lyu, Bowei He, Haolun Wu, Jikun Kang, Xue Liu. *QUACK: Questioning, Understanding, and Auditing Communicated Knowledge in Multimodal Social Deduction Agents*. In **EMNLP 2026**.

Weixu Zhang, Ye Yuan, **Changjiang Han**, Yuxing Tian, Zipeng Sun, Linfeng Du, Jikun Kang, Hong Kang, Xue Liu, Haolun Wu. *Preference Heads in Large Language Models: A Mechanistic Framework for Interpretable*

Personalization. In **ACL 2026**.

Ye Yuan, Weien Li, Rui Song, Zeyu Li, Haochen Liu, Xiangyu Kong, Zixuan Dong, Linfeng Du, Zipeng Sun, Weixu Zhang, Jiaxin Huang, **Changjiang Han**, Yonghan Yang, Zichen Zhao, Xiuyuan Hu, Haolun Wu, Yankai Chen, Fengran Mo, Jikun Kang, Bowei He, Dawn Song, Philip S. Yu, Xue Liu. *Discrete Diffusion Models: A Unified Framework from Tokenization to Generation*. arXiv preprint, 2026.

Dun Yuan, Fuyuan Lyu, Ye Yuan, Weixu Zhang, Bowei He, Jiayi Geng, Linfeng Du, Zipeng Sun, Yankai Chen, **Changjiang Han**, Jikun Kang, Xi Chen, Haolun Wu, Xue Liu. *Beyond Message Passing: A Semantic View of Agent Communication Protocols*. arXiv preprint, 2026.

Experience

AMD Remote
Research Intern Nov. 2026 – Present
Open Source & AI, AMD. Mentor: Xunzhuo Liu. Working on vLLM Semantic Router, the vLLM project's system that routes each LLM request to the most suitable model, an open-source project with 6.4K GitHub stars and adopted by Microsoft, Red Hat and Dell. Also working on the application and development of its Decision models, a family of open foundation decision models trained and served on AMD Instinct GPUs.

JD.com Beijing, China
Research Intern, Tech Genius Team (TGT) Apr. 2026 – Sept. 2026
Smart Marketing and Recommendation Department, JD Retail. Mentor: Qingyang Li. Worked on the OxygenREC generative recommendation architecture, including reasoning instructions for end-to-end generative recommendation. Co-first author of OxygenREC-v2; first author of SID-NL.

State Key Laboratory of Brain-Machine Intelligence, Zhejiang University Hangzhou, China
Research Intern May 2024 – Sept. 2024
Supervisor: Prof. Gang Pan. Worked on EEG data preprocessing, neuron parameter design for spiking neural networks, and hybrid SNN–ANN architectures.

Automated Chemistry Laboratory, Dian Group, HUST Wuhan, China
Project Leader June 2023 – June 2025
Led a two-year project building an AI-scheduled, vision-guided dual-arm robotic platform for automated chemical preparation, from design to deployment; designed the robot control.

Awards

Outstanding Undergraduate Thesis Award, HUST (Top 3%) 2025
Outstanding Graduate Award, HUST (Top 15%) 2025
First Prize (Hubei Province), China Undergraduate Mathematical Contest in Modeling (Top 10%) 2023
Honorable Mention, Mathematical Contest in Modeling (MCM) (Top 15%) 2022

Skills

Programming: Python, C++, Assembly; PyTorch, ROS.
Parallel Computing: multi-node distributed training on GPU clusters (32× A100 / B200).
Hardware & Communications: embedded circuit design; wireless communication.
Languages: Mandarin Chinese (native); English (fluent).

References

Prof. Xue (Steve) Liu (M.Sc. Supervisor)
Professor, MBZUAI and McGill University
FCAE, FIEEE

Qingyang Li (Internship Mentor)
Senior Research Scientist, JD Retail, JD.com